

Overview of Cross entropy loss

Subject: Cross entropy loss

$$H(p, q) = -\sum_x p(x) \log q(x)$$

1.

$$\frac{\partial L}{\partial w_j} = -\sum_{i=1}^N x_{ij} [y_i (1 - y_i) - (1 - y_i) y_i] = \sum_{i=1}^N x_{ij} (y_i - y_i^2) .$$

2.

$$L(\theta; x) = \prod_i q(X = x_i) = \prod_x q(X = x)^{x_i} \approx \prod_x q(X = x)^N \cdot p(X = x) = \exp \left[\sum_i \log q(X = x_i) \right] \\ = \exp \left[\sum_x x_i \log q(X = x) \right] = \exp \left[\sum_x x_i N \cdot p(X = x) \log q(X = x) \right]$$

3.

Motivation In information theory, the Kraft–McMillan theorem establishes that any directly decodable coding scheme for coding a message to identify one value x_i out of a set of possibilities $\{x_1, \dots, x_n\}$ can be seen as representing an implicit probability distribution $q(x_i) = \left(\frac{1}{2}\right)^{\ell_i}$ over $\{x_1, \dots, x_n\}$, where ℓ_i is the length of the code for x_i in bits. Therefore, cross-entropy can be interpreted as the expected message-length per datum when a wrong distribution q is assumed while the data actually follows a distribution p . That is why the expectation is taken over the true probability distribution p and not q . Indeed the expected message-length under the true distribution p is

4.

Cross-entropy loss function and logistic regression Cross-entropy can be used to define a loss function in machine learning and optimization. Mao, Mohri, and Zhong (2023) give an extensive analysis of the properties of the family of cross-entropy loss functions in machine learning, including theoretical learning guarantees and extensions to adversarial learning. The true probability p_i is the true label, and the given distribution q_i is the predicted value of the current model. This is also known as the log loss (or logarithmic loss or logistic loss); the terms "log loss" and "cross-entropy loss" are used interchangeably. More specifically, consider a binary regression model which can be used to classify observations into two possible classes (often simply labelled 0 and 1). The output of the model for a given observation, given a vector of input features x , can be interpreted as a probability, which serves as the basis for classifying the observation. In logistic regression, the probability is modeled using the logistic function $g(z) = 1 / (1 + e^{-z})$ where z is some

12.

Estimation There are many situations where cross-entropy needs to be measured but the distribution of p is unknown. An example is language modeling, where a model is created based on a training set T , and then its cross-entropy is measured on a test set to assess how accurate the model is in predicting the test data. In this example, p is the true distribution of words in any corpus, and q is the distribution of words as predicted by the model. Since the true distribution is unknown, cross-entropy cannot be directly calculated. In these cases, an estimate of cross-entropy is calculated using the following formula: